Bio 202
Final Term Mcq's
From Past Papers
By Antagonist

1) Eicosanios that cause the inflammation or called pro inflammatory.
   a) Omega 3
   b) Omega 6
   c) omega 9
   d) All of these
2) What is the correct ratio of omega 3 and 6 in the diet?
   a) 1:3
   b) 1:4
   c) 1:5
   d) 1:8
3) Omega -3 fatty acids such as alpha-linolenic acid and its derivatives has effects on human health is
   a) Inflammatory
   b) anti inflammatory
   c) no effects
   d) all of these
4) Docahexonic Acid (DHA) is selectively incorporated into___
   a) Cell membrane
   b) Retinal membrane
   C) postsynaptic neuralmembrane
   d) Band C
5) Omega -3 fatty acids are beneficial particularly for diseases r)
   a) Cardiovascular
   b) Alzheimer's
   c) Cancer
   d) All of these
6) Monoenoic Acids (Oleic Acid) are possiblaythe mostNioatty acids present in
   a) Olive Oil
   b) Corn
   c) soya bean
   d)peanut
7) Polyunsaturated fatty acids contain _ Double bonds
   a) Single
   b) Double
   c) More than one
   d) tripple
8) Example of Polyunsaturated fatty acids are.....
   1. Linoleic Acid
   2. olenic Acid
   3. Arachidonic Acid
   4. All of these
9) how many double bonds are present in Arachidonic Acid
   a) Single
   b) Double
   c) Triple
   d)Tetrad
10) Arachidonic Acid synthesized from ____
   a) Linoleic Acid
   b) Alpha Linolenic Acid
   c) Palmatic Acid
   d) None of these

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Acid

11) ___ is a precursor of paracrine hormones called Eicosanoids

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a) Linoloic acid
b) Alpha-linolenic acid
c) Arachidonic Acid
d) All of these

12) Eicosanoids exer complex control over many bodily systems in
a) inflammation
b) b)immunity
c) c)messengers
d) All ofthese

13) Palmetoleic Acid belongs to which family?
a. Omega 3
b. Omega 6
c. Omega 7
d. omega 8

14) Fatty acids commonly found in body which are
a) Palmitic Acid
b) Stearic Acid
c) Oleic Acid

15) All of these Excess carbohydrates in the body are converted into
a) Palmitic Acid
b) Stearic Acid
a) Acid
c) None of these
d)

16. the no. of carbon atoms in stearic Acid are
a) 15
b) 16
a) 17
c) 18

17. most common saturated fatty acid found in Animals, plants and microorganisms.
a) Palmitic Acid
b) Stearic Acid
c) Oleic acid
d) None of these

18.Eicosanoids are derived from fatty acids
a) Omega 3
b) Omega 6
c) Omega 8
D) Omega 3 and 6

19. is the first fatty acid produced during fatty acid synthesis
a) Palmitic Acid
b) Stearic Acid
d) Oleic Acid
e) None of these

20. stearic acid are most abundant in animal fat up to
a) 25%

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b) 30%
c) 35%

21. which is the monounsaturated Fatty acid found naturally
a) Palmitic Acid
b) StearicíOrcid
c) Oleic Acid

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d) None of these

22. which fatty acid is highly abundant in membrane lipids
   a) Palmitic Acid
   b) tearic Acid
   c) Oleic Acid
   d) None of these

23. Glycerol and sphingosine are the type of most commonly found in lipids
   a. Esters
   b. Alcohols
   c. Fatty Acids
   d. Amides

24.glycerol contains_____Carbon and ____ hydroxyl OH groups
   a. 3 and 3
   b. 3 and 4
   c. 2 and 3
   d. 2 and 2

25. is synthesized from Dihydroxyaceton phosphate
   a. Alcohols
   b. Fatty Acids
   c. Glycerol
   d. sphingosine

26. Dihydroxyaceton phosphate is acted upon by _ enzymes to form glycerol
   a) one enzyme
   b) two enzymes
   c) three enzymes
   d) four enzymes

27. which enzyme remove the phosphate group from glycerol phosphate to become glycerol
   a) Glycerol phosphate dehydrogenase
   b) glycerol kinase
   c) phosphatease
   d) All of these

28. Glycerol is a precursor for synthesis of ____ and_____in the liver and adipose tissue
   a) Triacylglycerols and phospholipids
   b) diacylglycerols and glycolipids
   c) monoacylglycerols and phospholipids
   d) tryacylglycerols and glycolipids

29. when the body uses stored fat as a source of energy glycerol and ____ are released into blood stream
   a) Amino Acids
   b) Glucose
   c) Fatty acids
   d) Triacylglycerols

30. sphingsine is an_
   a) Poly Alcohol
   b) Amino Alcohol
   c) glyco Alcohol

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d) None of these

31. sphingosine is synthesized in the body in the form of____ _

a) Amide
b) Ceramide
c) newmideyride

32. serine and palmitoyl CoA condense to form a product ketosphiganinine that is____ to sphinganine

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a) Oxidised
b) Reduced
c) Hydrolysed
a) Condensed

33. sphingosine is formed via degradation of sphingosine in the _
a) Lysosomes
b) Golgyappratus
c) Neucl s
d) Cytoplasm

34. In Acrolein test we used_____acid for dehydration to detect glycerol
a) Nitric Acid
b) Phosphoric Acid c
c) sulphuric Acid
d) Hydrochloric Acid

35. glycerol combines with three molecule to form glycerol trinitrate
a) Nitric Acid
b) Phoshoric acid
c) Suphuric acid
d) None of these

36. which enzyme is used to hydrolyzed the Triacylglycerols
a) Protease
b) Glycerol kinase
a) phosphatease
b) lipase

37. There are _ significant advantages to using triacylglycerol as stored fuels.
a) One
b) two
c) three
d) four

38. The carbon atoms of fatty acids are more___ than those of sugars.
a) Oxidized
b) reduced
c) condensed
d) All of these

39. Triacylglycerol yields more than___ as much energy as the oxidation of carbohydrates.
a) Twice
b) thrice
c) four times
d) 5 times

40. Triacylglycerol are____ and therefore____ _
a) Hydrophilic - Hydrated
b) Hydrophilic -Dehydrated
c) Hydrophobic -Dehydrated
d) Hydrophobic -Hydrated

41. the organism that carries fat as fuel doesn't have to carry the extra weight of
a) Proteins

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b) water
c) carbohydrates
d) none of these

42. One gram of polysaccharide has ____ ___.,ram of water
a) One
b) Two

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c) three
d) four

43. Moderately obese people with ___ kg of triacylglycerol deposited in their adipocytes could meet their energy needs for months.
a) 5 to 10 kg
b) 10 to 15kg
c) 15 to 20 kg
d) 20 to 25 kg

44. the human body can store less than ___ energy supply in the form of glycogen
a) A day
a) 2 days
b) 3days
c) 4 Days

45. In some animals triacylglycerol storeunderthe skin serve as insulation against_
a) Low temperature
b) High temp
c) Moderate temperature
d) All of these

46. Physical properties of N- are
a) colourless
b) tasteless
c) odorless
d) All of these

47. the specific gravity of all fats is less than
a) one
b) two
c) three
d) Four

48. The emulsification of dietary fats in intestinal canal brought about by
a) Bile pigments
b) Bile juice
c) Bile salts
d) All of these

49. During the hydrolysis of Fats it yields_
a) fatty acids
b) Glycerol
c) Fatty acids and glycerol
d) cant be hydrolized

50. During the hydrolysis ofTriolein is converted into
a) Palmitic Acid
b) Stearic Acid
c) Oleic Acid
d) None of these

51. hydrolysis of a fat by an_____is called saponification
a) Acid
b) Base

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c) Alkali
d) All of these

52. the alkali salts of the fatty acids which are called _
a) soaps
b) toothpaste
c) Detergents

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d) All of these

53. which Alkali is used to form sodium plamitate soap formation
   a) NaOH
   b) CaOH
   c) KOH
   d) NH4 OH

54. the amount of alkali needed to saponify a given quantity of fat will depend upon the number of_group present
   a) Amino group
   b) alkyl group
   c) carboxyl group
   a) group

55. Fats containing _ chain fatty acids will have _ COOH groups per gram than long chain fatty acids.
   a) Short-more
   b) long -more
   c) short -less
   d) long -less

56. saponification number of butter is
   a) 220 to 230
   b) 245 to 262
   c) 1 to 200
   a) d)210 to 233

57. Additive reactions consist of CD
a) Hydrogenation A
b) Haloge
c) A and B
d) Decomposition

58. During the hydrogenation OF lipid the melting point _and fatbecomes more_ atroom temperature.
   a) decrease -liquid
   b) Increase-Solid
   c) Decrease-Solid
   d) increase -liquid

59. During hydrogenation of fatty acids catalyst is used
   a) nickel
   b) copper
   c) Nickel and copper
   d) Zinc

60. Margarine manufacturing means_
   a) changing liquid to solid
   b) Solid to liquid
   c) Liquid to liquid
   d) All of these

61. Addition of halogens in unsaturated fatty acids is called
   a) Hydrogenation
   b) halogenation
   c) hydrohalogenation

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d) All of these

62. the degree of unsaturation is reflected by

a) Iodine number
b) sponification number
c) A and B
d) Sodium Number

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63. Iodine number is defined as the number of grams of iodine absorbed by _ of fat
    a) S0gm
    b) lO0gm
    c) 75gm
    d) 85gm
64. during the conversation of Linoleic acid at room temperature into Tetraiodostearic acid which catalysts are/is used?
    a) Acetic Acid
    b) Methanol
    c) Acetic Acid and Methanol
    d) None,c,
65. fats rich in unsaturated fatty acids have_
    a) High Iodine number
    b) Low iodine number
    c) Moderate number
    d) all of these
66. the higher is the iodine number the_ reactive and _ stable
    a) More- less
    b) Less- more
    c) More -More
    d) Less- more
67. Rancidity occurs particularly when expose to
    a) Light
    b) Heat
    c) Atmospheric
    d) All of these
68. ____ fats resist rancidity more than___ fats that have unsaturated double bonds
    1. Saturated-unsaturated
    2. Unsaturated-Saturated
    3. less saturated-
    4. More Unsaturated
69. Rancidity is occurs due to
    1. Oxidation
    2. Hydrolysis
    3. Hydration
    4. Oxidation and Hydrolysis
70. oxidation of the fatesmoecule give rise to some short chains of
    e) Aldehydes
    f) ketones
    g) dicarboxylic acids
    h) All of these
71. The Rancidity be prevented in the food by adding
    a) Vitamin E
    b) Vitamin D
    c) Vitamin A
    d) All of these

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72. Second type of neutral lipids that are of physiological importance.
    a) Glycerol
    b) alcohol
    c) waxes
    d) aldyhides
73. waxes are___ in water and in fat

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a) Soluble - insoluble
b) Insoluble-soluble
c) soluble -soluble
a) d)insoluble-insoluble

74. waxes are _ by lipases
a) Digestible
b) indigestible
c) semi digestible
d) no effect

75. there are_ types of waxes
a) One
b) two
c) Three
d) Four

76. True waxes are solid simple lipids containing a
a) Monohydric Alcohol
b) Dihydric alcohol
c) trihydric alcohol
d) polyhydrigiKhol

77. Which alcohol compound is found in plant cuticle waxes and in bees wax
a) MyricylAlohol
b) CetylAlohol
c) Polyhydric Alocohol
d) A and B

78. Example of true waxes is/are
a) Bee wax
b) spermaceti wax
c) Cholesterol
d) A and B

79. Bees wax is a mixture of waxes its chief constituent is
a) Myricyl and Palmitate
b) Myrical and Aceatic Acid
c) Palmitate and Oleic Acid

80. sperm whale contains wax in his head
1. 330Kg
2. 300Kg
3. 3N9Kg
4. 336kg

81. the wax obtained by Brazilian palm tree is_
a) Bees was
b) Camauba wax
c) jojoba wax
d) All of these

82. waxes used for candles, soaps and cosmetics is
a) Bees was
b) CarriltilAwa-x

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c) jojoba wax
d) All of these

83. LDLcholestrol and HDL cholesterol stands for?
no options
a) Low density Lipoprotein
High density lipoprotein

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84. which type of cholesterol have been associated with decreased risk for heart diseases
    a) LDL cholesterol
    b) HDL cholesterol
    c) TAG
    d) Total cholesterol
85. lipoprotein particles are spherical aggregates with ____ lipids at the core and _ protein side chains and lipid head groups at the surface
    1. Hydrophobic-hydrophilic
    2. Hydrophilic -hydrophobic
    3. Hydrophobic -Hydrophobic
    4. Hydrophilic -Hydrophilic
86. Lipoproteins function is to keep their components lipids as
    a) Soluble
    b) Insoluble
    c) Semisoluble
    d) None of these
87. Different types of lipoproteins are different in
    a) Size
    b) Density
    c) Site of origin
    d) All of these
88. The maximum concentration of TAG (Triacylglycerol) present in
    a) Chylomicrons
    b) VLDL
    a) LDL
    b) HDL.
89. the chylomicron remnet is removed from the circulation by the _
    a) Kidney
    b) Intestine
    c) Liver
    d) testinal Mucosa
91. are assembled in the liver
    a) Chylomicron
    b) VLDL
    c) HDL
92. LDL contain___ as their major apolipoprotein that carry cholesterol to_
    a) apoB-100 and Extra Hepatic Tissue
    b) apoB-200 and Liver
    c) ApoB-100 and liver
    d) ApoB-200 and Extra hepatic Tissue
93. ____ can accumulate in the_ lining the arteries resulting tin the formation of atherosclerosis.
    a) Reduced LDL and Macrophage cells
    b) Oxidized LDL and Macrophage cells
    c) Oxidized HDLand macrophage cells
    d) Reduced HDL and Macrophage cells
94. which type of lipoproteins originates in the liver and small intestine

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a) Chylomicron
b) VLDL
c) LDL
d) HDL

95. which type of lipoprotein take up cholesterol from peripheral tissues and return it to the liver as cholesterol esters

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a) Chylomicron
b) VLDL
c) LDL
d) HDL

96. when cholesterol is taken up by the HDL, it is immediately _ and becomes_
a) Esterified - Hydrophobic
b) Oxidized - Hydrophobic
c) Esterified - Hydrophilic
d) Oxidized -Hydrophilic

97. when cholesterol is taken up by the HDL it is called _
a) Nascent HDL
b) Mature HDL
c) Pre Mature HDL
d) Atikthese

98. which lipoprotein is called good cholesterol carrier
a) Chylomicron
b) VLDL
c) LDL
d) HDL

99. Healthy Fats contain _
a) Monounsaturated fats
b) polyunsaturated fats
c) omegasaccrides
d) All of these

100. Saturated aft is assocated with levels of total cholesterol and LDL cholestrol
a) Low
b) High
c) Medium
d) Very low

101.. Prostaglandins are fatty acids containing atoms, including an internal
a) 10 C- 5 carbon ring
b) ISC -8 carbon ring
c) 20 C- 5 Carbon Ring
d) 20 C - 8 carbon ring

102. Prostaglandins have double bond between
a) C13andC14
b) CS and C6
c) C17 and C18
d) All of these

103. Nomenclature of prostaglandins (PGs) involves assignment of
a) Capital letter
b) Numeral subscript
c) Greek letter subscript
d) All of these

104. Nomenclature of (PGs) the capital letter refers
a) Ring OH group

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b) Ring Double Bond
c) Ring substituents
d) All of these

105. Which prostagland ins contain 2 rings.,,,,,.•:
a) PGA
b) PGD

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c) PGI
d) All of these

106. The no of unsaturated bonds in Prostaglandins is determined by
a) Capital letter
b) Numeral subscript
c) Greek letter subscript
d) All of these

107. is there are two double bonds present in prostaglandins is determined
a) Series one
b) Series two
c) Series three
d) Series four

108. The Greek letter subscript found only in series it refresh group at carbon 9
a) F - OH group
b) A - OH group
c) F -COOH
d) A - COOH group

saturated fats are found in
a) Butter and cheeses
b) milk and palm oil
c) Animal fat and coconut oil
d) all of these

109. Among the SFAs stearic acid appears to have effect on LDL-C
a) Negative
b) Neutral
a) Positive
b) Polar

110. Saturated Fatty Acids (SFAs) are considered to Hypercholesterolemic
a) Luaric Acid
b) Myristic
c) palmitic
d) All of these

111. Denary cholesterol has on plasma cholesterol go
a) Little effect
b) High Effect
c) Medium Effect
d) low effect

112. There are classes of phospholipids
a) one
b) two
c) three
d) four

113. Membrane lipids are i.e one end of the mol le is hydrophobic and the other ishydrophilic-
a) Amphoteric
b) amphipathic
a) tholhatic

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c) none of these

114. Phospholipids is not synthesized c
a) Liver
b) adipose tilt!frv
c) mammary glands
d) mature erythrocytes

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115. Phosphoglycerideisakacylgtycerolwith a group on the carbon of glycerol

a) phosphate - 3 carbon \
b) Amide - 3 carbon
c) sulphate - 3 carbon
d) phosphate - 5

116. in general glycecophospholipids contain at Carbon no 1(C-1)
a) Saturated
b) Unsaturated
c) Phosphate
d) All of these

117. the phos). ate group on phophatidic acid PA can be esterified to another compound containing
a) Amide group
b) Alkyl Group
c) Hydroxyl group
d) Ester group

118. the most abundant phospholipids of the cell membrane is
a) Lecithins
b) Phospholipidcholine
c) phospholipidac,
a) d)cetaline

119. the gaseous exchange occurs in lungs
a) Alveoli
b) bronchioles
c) alveolus
d) All of these

120. Adenosine 3'-phosphate-5'-phosphosulfate is the sulfate donor for:
A) sulfated proteoglycans
B) Conjugate of drugs
C) Phosphosulfate
D) Both a and b

121. Which of the following methyl group donor:
A) S-adenosylmethionine
B) Adenosine
b) Phosphate -5
c) All of these

122. and play key roles in activating or inhibiting proteins:
a) GTP, GMP
b) GDP, AMP
c) GTP, GDP
d) AMI:VMP

123. cAMP and cGMP serve as messengers in signal transduction pathways: (ZoNa)
1st
b) 2nd
a) c)3rd
c) 4th

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124. synthetic purine and pyrimidine analogs contains which group?
    a) Halogens
    b) Thiols
    c) additional of nitrogen atoms
    d) All of these
125. Pyrimidine are used for:

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a) Chemotherapy
b) Treatment of AIDS
c) Organ translation
d) All of these

126. suffix "me" in bases denotes the presence of in the ring.
a) Nitrogen
b) Amine
c) Vlroxyl
d) Both a and b

127. The nitrogen -containing bases belong to families of cornet4:
a) 2
b) 3
c) 4
d) 5

128. By the attachment of different groups to the different typlrs of pyrimidine and purine are generated.
a) Rings
b) Compound,
c) Nucleotide
d) Both a and b

129. which nucleotide contains AG:
a. DNA
b. RNA
c. Both
d. None of these

130Adenine when combines with the structur-eis known as Adenosine
a) Hexose
b) Pentose
c) Tetrose
d) All of these

131. Deoxyguanosine refers to which base?
A) Adinine+ pentose
b) Guanine+pentose
c) Guanine+ribose
d) Guanine+sucrose

132. Which of the following is Minor Purine Bases:
a) Inosine
b) methyl guanine
c) Both a and b
d) Guanine

133. Mercaptopurine, Allopurinol & 8-Azaguanine are referred to which type of bases?
a) Natural
b) Unnatural
c) Doubles
d) Both a and b

134. Uric acid is the catabolic end product of in human beings:

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a) Guanine'
b) Purine
c) thymine
d) Cytosine

135. in the prostaglandins the Hydroxyl group OH primarily exists - lh Position
a) Alpha

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b) Beta
c) Gamma
d) Omega

136. Act as substrate for the lipooxygenase
a) Arachidonic Acid
b) 20 C Fatty Acids
c) Arachidonic Acid and 20 C Fatty Acids
d) Arachidonic Acid and 15 C FAs

137. Products of Lipoxygenase pathway are
a) Cyclic
b) Linear
c) Cyclic or linear
d) double ringed

138. Nomenclature rules are followed for Leukotriene and lipoxins starts from
a) from series 1
b) from series 2
c) from series 3
d) All of these

139. ETA, and EPA stands for no options
Eicosa tri enoicAcidEicosaPentaenoic Acid

140. Arachidonic acids have double bonds
a) one
b) two
c) three
d) four

141. lipoxygena se enzymes incorporation of_ molecule in arachidonic acid and forms
a) Oxygen - OOH
b) Oxygen -COOH
c) Oxygen - OH
d) Hydrogen - OH

142. when oxygen is added it isomerizes at away from the hydro-peroxy
a) One carbon
b) Two carbon
c) Three carbon
d) Four carbon

143. addition of oxygen during synthesis of leukatrienes and lipoxins configuration changes to
a) Cis to trans
b) Trans to Cis
c) Cis to double trans
d) Tranato Double Cis

144. During the isomerization of double bond becomes in Lipoxygenase pathway
a) one step forward
b) one step backward
c) two step forward
d) step backward

145. The Hydro-Peroxy group is cc\Noor

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a) Stable
b) Unstable
c) Very stable
d) low stable

156. which of the following molecules act as messenger moleculese
a) HETEs

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b) Archonadic Acid
c) HerV
d) All of these

157. LTs and D(s are stands for no option
Leukotrienses and Lipoxins
HETEs: Hydroxy-eicosa-tetra-enoic _acid
HPETES: Hydro-peroxy-eicosa-tetra-enoic acid

158. LTs and D(s have double..... as campared to Prostaglandins have double bonds
a) 1 to 3 and 3 to 5
b) 3to5andlto3
c) 2 to 5 and 3 to 5
d) 2 to 4 and 2 to 5

159. HPETEs are the precursors of
a) leukotrienes 4
b) lipoxins
c) leukotrienes and lipoxins
d) HETEs

160. 5-HPET is converted to
a) LTA4
b) LTA5
c) LTA6
d) LTA7

161. Functionalleukotrienes are formed from LTA4 examples are
a) LTB4
b) LTC4
c) LTD4
d) A and B

162. Removal of residue from ITC4 forms LTD4
a) Glycine
b) Glutamine
B) Glutamate
c) Band C

163. Leukotrienes were so named because they were first discovered in
a) Red blood cells
b) White blood cells
c) Platelets
d) All blood cells

164. The lipoxins are formed through the action of and on Arachidonic Acid
a) 15 and 5 lipoxygenase
b) 10 and 5 lipoxygenase
c) 5 and 5 lipoxygenase
d) All of these

167. Chemotaxis is migration of _to the tissue site of injury or inflammation
a) RBCs
b) WBCs
c) Platelet

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d) All of these

168. Lipoxins induce chemotaxis and stimulate radicals for killing microorganisms
a) Oxide
b) Super oxide
c) Trioxide

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d) All of these

169. which of the followings have very short half lives
a) Prostaglandins
b) Thromboxanes
c) leukatrienOwectjjeoxins
d) All of these

170. Nucleotides are the building blocks of .No optio0

Ans is Nucleic Acid

178. Nucleic acids, which include DNA and RNA,arnVre from monomers known as
a) Nucleoside
b) Nucleotides
c) Poly nucleotides
d) All of these

179. Protein synthesis dependent on
a) Nucleoside
b) Nucleotides
c) Poly nucleotides
d) All of these

180. Principal donors and acceptors of phosphoryl group in metabolism
a) ATP
b) ADP
c) AMP
d) ATP and ADP

181. in ATP molecules the removal of Group release energy
a) Sugar group
b) Ribose group
c) Phosphate
d) All of these

182. Examples of Co- Enzymes are
a) coenzyme A,
b) FAD
c) NAD+and NADP+
d) All of these

183,Because of reversibility of reactions, conversion of product su rate:
a) Forward
b) Back
c) Stop at
d) Both a and b in some cases

184) One simplifying approach in kinetics experiment to measure the:
a) Initial rate
b) Designated VO
c) 1.51greater >concentration
d) All of these

185) VO is the
A) Initial velocity
B) Final velocity'

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186) Increase in substrate_ reaction

A) Increase
B) Decrease
a) Stop
b) Both band c

187) When initial velocity (Vo) is plotted against [S] a hyperbolic curve results, which represent the :

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a) Minimum velocity
b) Maximum velocity
c) Substrate concentration
d) Initial velocity

188) At relatively low concentrations of substrate, Vo almost linearly:
a) Increases
b) Decreases
c) Go forward
d) Go reverse

189) At this point in the reaction, if [S] » E, all available enzyme is
a) Unsaturated
b) Saturated
c) Polymerase enzyme
d) Helicase enzyme

190) finite limit of is called saturation kinetics
a) [S]
b) Vmin
c) E
d) Vmax

191) Michaelis constant is represented by? C'ol31
a) Mc
b) Mk ISC481
c) Km
d) Mhc

192) Low concentration of substrate is needed to half -saturate the enzyme i.e. to reach a velocity that is?
a) IVMax
b) 1/2Vmax
c) 2Vmax
d) 2/3Vmax

193) A numerically Km reflects a affinity of e
a) High, high
b) High, low
c) Low, low.
d) Low, high.

194) Km glucose of approximately
a) 0.5mM
b) 0.05 mM
c) 0.005mM
d) 5mM

195) Write Michaelis-eenten equation, the rate equation with its symbols detail;..... no option
Ans is
Vo= vmax[S]/krn+[S]
Vo =Vmax [S]
Km+ [S] Where;
• Vo= initial reaction velocity
• Vmax = maximal velocity

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•Km= Michaelis constant
• = Kl+K2 / K-1
• [S] = substrate concentration

196) Leonor Michaelis and Maud Menten in proposed a simple model that accounts for most of the features of enzyme -catalyzed reactions:

a) 1919

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b) 1911
c) 1939
d) 1913

197) What is abbreviation of ES?
a) Enzyme study
b) Enzyme starting
c) Energy substrate
d) Enzymes substrate

198. is the most abundant substance in living systems, making up% or more of the weialk Z o ost organisms
a) Water - 80%
b) Protein - 70%
c) Water - 70%
d) Carbohydrate - 70%

199. Are of crucial importance to the structure and function of biomolecules
a) Hydrogen bonding
b) Polarity
c) Tendency of water to ionize
d) A and C

200. What gives water to its unusual properties like higher NO:13Pand 'Heat of Vaporization
a) Covalent bonding
b) Hydrogen bonding
c) Dipole forces Intermolecular forces

201) Ether lipids compounds constitute as much as of the phospholipids of brain and muscle
a) 5%
b) 10%
c) 15%
d) 20%

202) Ether lipids with an unsaturated group (alkenyl) at the 1st position on the glycerol chain are called :
A) Plasmalogens
b) Glycerophospholipids
c) (PAF) Platelet -activating factor

203) PAF is synthesized and released by a variety of cell:
a) Size
b) Types
c) Length by numbers
d) Both a and b

204) About of the heart phospholipids are plasma logens
a) All
b) Half
c) One fourth
d) Some

205) How many subclasses are There of sphingolipids:
a) 2
b) 3
c) 4
d) 5

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206) Sphingoglycolipids molecules contain and components:

a) Carbohydrate
b) Lipids
c) Both a and b
a) None of these

207) Sphingolipids, like other membrane lipids, are compcis

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a) hydrophobic portion
b) polar head _grou.p
c) Plasmogens
d) BOTH A AND B

208) Sphingomyelins are present in the animal cells:
a) plasma membranes
b) Nucleus
c) Cytoplasm
d) All body

209) Cerebroside have ir' Nar)pfked to ceramide:

a) single
b) Double
c) Triple
d) Botha nd d

210) Galactosylceramide is a major glycosphingolipid of
a) Nucleus
b) Digestive system
c) Brain
d) Both a and b

211) Globosides are •
a) Glycosphingolipids
b) Galactosylceramide
c) Cerebrosides
d) Both a and b

212) Globosides are highly abundant in
a) RBCs
b) Wbc
c) Platelets
d) All of these

213) Sialic acids are acidic sugars with a carbon backbone:
a) Six
b) Seven
c) Eight
d) Nine

214) Sialic acid gives gangliosides the charge at pH 7
a) Neutral
b) Positive
c) Negative
d) Both band c

215) About of brain lipids are gangliosides and were first isolatfffrbtjhe ganglion of brain cells
a) 4%
b) 5%
c) 6%
d) 7%

216) Gangliosides have a significant role in

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a) Growth
b) differentiation of tissues
c) carcinogenesis
d) All of these

217) The basis of blood groups depeorrcoreiggron of different antigens on
a) RBCs

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b) Wbc
c) Platelets
d) All of these

218) Modification ofO afititenreSults in blood groups respectively:
a) A
b) B
c) AB
d) All of these

219) Sulfatides are found predominantly in?
a) nerve tissue
b) Kidney
c) Stomach
d) Both a and b

220) Gangliosides have oligosaccharides as their head groups:
a) Non_polar
b) Polar
c) Neutral
d) Both a and b

221) Glyceroglycolipids are the predominant membrane lipids of?
a) Animals
b) Plants
c) Prokaryots
d) Eukaryotes

222) Galactolipids constitute about of plant membrane lipids:
a) 50% to 60%
b) 60%to 70%
c) 70%to80%
d) 80%to 90%

223) abundant lipids in the biosphere are?
a) Sulfolipids
b) Galactolipids
c) Glycolipids
d) Glyceroglycolipids

224) Sulfonated glucose is joined to the of diacylglycerol in glycosickage:
a) C2
b) C3
c) C4
d) CS

225) Galactolipids and Sulfolipids are included in?
a) Glyceroglycolipids
b) Glycolipids
c) Glycerolipids
d) None of these CDtc

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226) 0 antigen also referred to as i gen:

B) A
C) B
D) C
E) H

227. Nucleotides that includes in pyrimidine are/is

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a) Cytosine
b) thymine
c) Uracil
d) All of these

228. Cytosine when combine with pentose it be
a) deoxyctytidine
b) cytidine
c) monoxycytidine
d) A&B

229. Which of the following is 2.4-dioxy-5-Methyl-Pyrimidine?
a) Cytosine
a) Thymine
b) Uracil
c) Adenine

230. 'Nit is the structure ofx eis
a) Pyrimidine
b) Purine
c) Cytosine
d) None

231. the difference between thymine and Uracil is
a) Nitrogen group
b) Hydroxyl Grott
C) Alcohol Group
c) Methyal Group

232. Which of the following is Minor PyramidineBase
a) Fluorouracil
b) ceytosine
c) Dihydrouridine'
d) Non of these

233. DHU, SFU and AZC stands for
DHU - Dihydrouridine SFU - Flouro uracil

AZC- Aza Cytosine
I

234. Which of the followingnucleotide can b degraded in human cells

a) Purine
b) Pyrimidine
c) ATP
d) RNA

235. During the stability of B-Alanine and B-Amino-iso-butyrate the production is
a) NH3
b) CO2
c) 02
d) A&B

236. sugar that found in nucleic Acids

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a) D- Ribose
b) 2-Deoxy D- Ribose
D) L- Ribose
c) A&B

237. the difference between Dexyribose and Ribose is
a) Addition of Hydrogen at C- 2'

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b) Addition of Oxygen at C-2'
c) Addition of hydroxyl group at C-3'
d) Addition of Hydroxyl group at C-2'

238. Ribose and Deoxyribose are belongs to
a) Bfamily
b) C Family
c) D family
d) All of these

239. Ribose and Deoxyribose present iasFuranose (ring) in the form of
a) B-Anomer
b) D-Ante2
c) Aldo-Anomer
d) Pento-Anwer

240. The addition of a pentose sugar to a bas produces a
a) Nucleotide
b) Nucleoside
c) Nucleoprotein
d) Nucleosome

241. Nucleotides are

E) Easter of Methyl
F) Easter of Nucleosides
G) Easter of purine
H) Easter of Pyrimidine

242. Which of the folio lug Energy molecules contain more energy
a) AMP
b) ADP
c) ATP
d) GMP

243. Phosphate groups give an overall charge to the nucleotides
a) Positive
b) Negative
c) Neutral
d) All of these

244. The nitrogen containing bases are i.e they have alternate bonds
a) Aliphatic - double
b) Aromatic -Tripple
c) Aromatic - Double
d) Aliphatic - Tripple

245. structure that contain other atoms in addition to carbon in ring structure called
a) Homocyclic
b) Hetrocyclic
c) Isocyclic
d) polycyclic

246. Nucleotides have nature

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C) Aromatic
b) Hetrocyclic
c) Aromatic and Hetrocyclic
d) Homocyclic

247. the six atoms rings of Purine and pyrimidines are numbered in
a) Same direction

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b) Opposite direction
c) Linear Direction
d) Cyclic Direction

248. which of the nucleoside contain double ring
a) Purine
b) Pyrimidine
c) Cycline
d) Hexocycline

249. Purines or pyrimidineswit an J are weak bases
a) Amino Group
b) HydroxylGroup
c) Methyl group
d) easter group

250. the mostirniTtnt functional groups of pyrimidines and purinse are no options
AnsNitrogerkRe>Cybonyl groups Exocylci Amino Group

251. Purine Ad Pyrimidine bases are
a) Hydrophilic
b) Hydrophobic
c) Neutral
d) All of these

252. The conjugated double bonds of purine and pyrimidine derivatives absorb
a) Infra red Light
b) Heat Waves
c) UV Light
d) all of these

253. The mutagen property is utilized in analysis of nucleotides and nucleic Acids
a) Qualitative Analysis
b) Quantitative Analysis
c) Qualitative and Quantitative Analysis
d) Numeric Analysis

254. purine and pyrimidine bases can exist in form at physiologic pH
a) keto-enol
b) amine-imine
c) A&B
d) Aminol

255. Smaller pyrimidine molecule has the and the larger purine molecu
a) Longer name - shorter name
b) Shorter name - Large name
c) Longer name - longer name
d) Shorter npie -Shorter name

256. Sugars are linked to the heterocyclic by a
a) Glycosidic Bond
b) N- Glycbsidic Bond
D) B-N-GlycosidicBond,
c) All of these

257. B-N-Glycosidic Bond iftyrimickne forms at and Purine at

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a) N-1 and N-9
b) N-3 and N-9
c) N-1 and N-3
d) At any position

258. Glycosidebond is formed when removed from pentose and_ removed from Base
a) OH and H

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b) Hand OH
c) OH and 0
d) 0 and H

259. The difference between glycosidic linkage and 0-glycosidic bond is
a) Nitrogen and Oxygen
b) Carbon and Nitrogen
c) oxygen and Hydroxyl
d) Amino and Carboxyl

260. The atoms in the rings of the bases are numbered in pyrimidines& in purines
a) 1 to Sand 1 to 6
b) to 6 and 9
a) 1 to 9 and 1 to 6
b) to7and9

261. in the structure of Nucleotides the prime no is given to
a) Nitrogen Base Carbons
b) Sugar carbons
c) Phosphate
d) All of these

262. if we say 5'- carbon in a nucleoside or nucleotide is referred to

a) Nitrogen Base Carbons
b) Sugar carbons
c) Phosphate
d) All of these

263. When two or more nucleotides combine together a bond formed
a) Glycosidic Bond
b) Peptid
c) phosphodiester Bond
d) Hydrogen Bond

264. phosphodiester bond is formed between group of sugar and group of other base
a) 3' OH and 5' PO4
b) 2' OH and 5' PO4
c) 5' OH and 3' PO4
d) 5' OH and 5sPO4

265. The backbones of both DNA and RNA
a) Hydrophobic
b) Hydrophilic
c) Neutral
d) None of these

266. The formation Of Hydrogen bonding in nucleotides is because
a) OH group of Sugar
b) Amino Group of Bases
c) Phosphate negative charge
d) All of these

267. The phosphate group completely ionized at pH
a) Acidic pH

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b) Basic pH
c) Neutral pH
d) All of these

268. The negative charges on phosphate are generally neutralized by with positive charges
a) Ionic interactions
b) Covalent Interaction

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c) Polar interactions
d) Non polar interactions

269. Cyclic G MP serves as second messenger in response to
a) Super oxide
b) Nitric oxide
c) Super Hydroxide
d) All of these

270. DNA present in
a) Chromosomes of Eukaryotes
b) Mitochondria
c) Chloroplast and plasmids
d) All of these

271. DNA is a polymer of
a) Deoxyribonucleoside monophosphate
a) b Deoxyribonucleosidediphosphate
b) Deoxyribonucleoside Triphosphate
c) All of these

272. The protein -DNA complex is present(anon membrane bound region known as
a) Plasmid
b) Nucloside
c) Nucleoid
d) Nucleolus

273. In DNA double helix, the two strands of DNA are held together by.
a) glycosidic linkage
b) Phosphodiester linkage
c) Hydrogen bonding
d) other interactions

274. The complex folding of large chromosomes within eukaryotic chromatin and bacterial nucleoids is generally considered
a) Primary structure
b) Secondary structure
c) tertiary structure
d) All of these

275. The stacking also involves a combination of interactions between the bases.
a) Van der Waa ls
b) Dipole - Dipole
c) Hydrogen bondings
c) A&B

276. Base stacking helps to _contact of the bases with water
a) Minimize
b) Maximize
c) no effect
d) All of these

277. A key factor affecting the rate of a reaction catalyzed by an enzyme is the conce45 of;
a) Enzyme
b) substrate

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c) pH
d) Temperature

178. The water molecules have partial charges
a) positive
b) Negative
c) One negative and 2 positive

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d) All of these

179. Attraction between Oxygen atom of one water molecule and hydrogen of another molecule this bonding is called
a) Covalent ballifirtf
b) Electrostatic forces
c) Hydrogen bonding
d) All of these

280. Oxygen atom allows each water molecule to form hydrogen bonds with as many as neighboring water molecules
a) One
b) Two
c) Three
d) Four

281. The degree of ionization of water into H+ and OH- ions is
a) Higher
b) Smaller
c) Mediator
d) None

282. During ionization of water the hydrogen ions hydrated to
a) Ammonium Ion
b) Hydronium Ion
c) Hydroxyl Ion
d) Hydronium Ion

283. The ionization of water can be measure by its
a) Head conductivity
b) Hydrogen conductivity
c) protonation
d) Electrical Conductivity

284. At 250C only about out of every IOpower 9 molecules in pure waterrikinized at any instant
a) one
b) Two
c) Three
d) Four

285. Write the equilibrium constatn for the reversible ionization of water..... NO option
Keq = [H+][OH-]/[H2O]

286. One mole of water is weighs is
a) 18g
b) 18 amu
c) 21g
d) 21 amu

287. The product in water equilibrium is designated as
a) Ke
b) Keq
c) Kw
d) keqw

288. The value for Keq determined by electrical conductivity measurements of pure water is at 250C

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a) I.8x 10 power-16
b) I.9x 10 power+16
c) I.8x10 power -18
d) I.9x10power -1

289. When there are exactly equal concentrations of H+ and OH-, as in pure water, the solution is said to be at_
a) Acidic pH

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b) Basic pH
c) Alkali pH
d) Neutral pH

290. PH means
a) Negative logarithm
b) Negative logarithm of Hydrogen ions
c) Negative logarithm of HO ions
d) Positive logarithm of Hydrogen

291. For each pH unit less than 7.0, the [H+] is tenfold and for each PH unit above 7.0, it is _tenfold
A) increased and decreased
B) Decreased and increased
C) less and decreased
D) more and increased

292. pH + p0H =
a) 11
b) 12
c) 14

293. What will be the pH ofO.l N CD
a) 1
b) -1
c) +l
d) 2

294 The stronger the acid, the greaerits tendency ionizes that means...
a) to gain electrons
b) to gain proton
c) to lose its proton
d) to gain electrons

295. Which of the following acids are strong acids?
a) Sulphuric Acid
b) Hydrochloric Acid
c) Nitric Acid
d) All of these

296. Strong bases are/is
a) NaOH
b) CaOH
c) NH4OH
d) All of these

297. The stronger the tendency to dissociate a proton, the stronger is the acid and the itspKa
a) stronger
b) higher
c) neutral
d) lower

298. is used to determine the amount of an acid in a given solution
a) Molarity
b) pH
c) Titration

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d) normality

299. A measured volume of the acid is titrated with a solution of known concentration

a) weak Base
b) Strong Bas
c) neutral solution
d) All of these

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300. The pH of the equimolar solution of acetic acid and acetate is exactly equal to the pKa of acetic acid pKa=
   a) 4.32
   b) 4.75
   c) 4.76
   d) 9
301. The Henderson Hassebalch HH equation relates
   a) pH
   b) pka
   c) buffer concentration
   d) All of these
302. The protein -DNA complex is present(anon membrane bound region known as
   a) Plasmid
   b) Nucloside
   c) Nucleoid
   d) Nucleolus

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Bio 202

Final Term Mcq's

From Past Papers

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